START

CREATE database hospital_database

CREATE hospitals table

SET hospital_id as PRIMARY KEY (INT)

DEFINE name as VARCHAR(100)

DEFINE address as VARCHAR(255)

DEFINE size as INT

DEFINE type as VARCHAR(50)

DEFINE accreditation_status as VARHAR(50)

CHECK if table is created

IMPORT hospitals.csv into hospitals table

VERIFY records imported correctly

CREATE doctors table

SET person_id as PRIMARY KEY

SET name as VARCHAR(100)

DEFINE date_of_birth as DATE

DEFINE address as VARCHAR(255)

DEFINE role as VARCHAR(50)

SET hospital_id as FOREIGN KEY

LINK doctor’s hospital_id to to hospital’s hospital_id

CHECK if table is created

IMPORT doctors.csv into doctors table

VERIFY records imported correctly

CREATE patients table

SET person_id as PRIMARY KEY

SET name as VARCHAR(100)

DEFINE date_of_birth as DATE

DEFINE address as VARCHAR(255)

DEFINE role as VARCHAR(50)

SET doctor_id as FOREIGN KEY

LINK patient's doctor_id to doctor's person_id

CHECK if table is created

IMPORT patients.csv into patients table

VERIFY records imported correctly

CREATE prescriptions table

SET prescription_id as PRIMARY KEY

SET patient_id as FOREIGN KEY

SET doctor_id as FOREIGN KEY

DEFINE medication as VARCHAR(100)

DEFINE prescription_date as DATE

LINK prescription's person_id to to patient's person_id

LINK doctor_id to doctor's person_id

CHECK if table is created

IMPORT prescriptions.csv into prescriptions table

VERIFY records imported correctly

VERIFY all four tables have correct records

VERIFY foreign key relationships are valid

END

START hospital database queries

Query 1: Print list of all doctors at a hospital

INPUT hospital_id

RETRIEVE doctors linked to this hospital

DISPLAY doctor details

Query 2: Print list of all prescriptions for a patient, ordered by prescription date

INPUT patient_id

RETRIEVE prescriptions linked to this patient_id

SORT prescriptions by prescription date

DISPLAY prescriptions in date order

Query 3: Print list of all prescriptions that a particular doctor prescribed

INPUT doctor_id

RETREIVE prescriptions linked to doctor_id

DISPLAY prescriptions

Query 4: Add a new patient to database, including being registered with one of the doctors

INPUT new patient details

INPUT doctor_id

CHECK IF doctor_id exists in doctor table

IF doctor exists THEN

  INSERT new patient with assigned doctor_id

ELSE

  DISPLAY error message

END IF

CONFIRM patient is added

Query 5: Identify which doctor made most prescriptions

COUNT prescriptions for each doctor_id

FIND highest prescription count

RETREIVE doctor and prescription total

DISPLAY result

Query 6: Print list of all doctors at the hospital with biggest size (number of beds)

FIND hospital with highest number of beds

RETRIEVE hospital_id of that hospital

RETRIEVE all doctors with that hospital_id

RETRIEVE doctor's names and hospital name

END